Mahmud Al Muhaimin

AI Engineer | Software Engineer

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PROFESSIONAL SUMMARY

AI-focused software engineer working across LLM systems, agentic workflows, machine learning and full-stack product development. Hands-on with RAG, AI agents, APIs, React/Next.js, Node.js and PyTorch/TensorFlow, from model/API integration through backend architecture and user-facing products. Has also delivered digital growth work for 248+ clients across SaaS, e-commerce and travel.

CORE SKILLS

AI / GenAI	LLM applications, RAG, AI agents, agentic workflows, embeddings, semantic search, AI orchestration
Programming	Python, TypeScript, JavaScript, SQL, C++
Backend	Node.js, Express, FastAPI, REST APIs, SQLite, MySQL
Frontend	React, Next.js, responsive UI
Machine Learning	PyTorch, TensorFlow, CNN, LSTM, VAE, computer vision, recommender systems
Engineering	Software architecture, testing, Git/GitHub, Linux, API integration, product prototyping

SELECTED PROJECTS

SuprBuild Mesh *AI Systems / Backend*

- Built an agent-oriented task mesh on Node.js, Express and SQLite handling agent coordination, task execution, reputation scoring and authenticated access.
- Supports 10+ concurrent agents across 20+ REST endpoints, with ~120 ms median API response time in local load tests.

Route52 *Bangla AI / Full Stack*

- Built a Bangla-native AI assistant in TypeScript and React with conversational workflows, authentication, persistent chat history and an agent-oriented architecture.
- Beta-tested with 30+ users across 200+ conversations to refine prompts and conversation flow.

PhyloVAE *Machine Learning / PyTorch*

- Implemented variational autoencoders in PyTorch for phylogenetic/tree-structured data, reducing reconstruction loss by ~18% versus a baseline autoencoder.

Quick-Draw Detection *Computer Vision / Deep Learning*

- Trained a CNN + Bi-directional LSTM classifier on a 20-class QuickDraw-style dataset (~100k images), reaching ~92% test accuracy, +6 points over a CNN-only baseline.

PROFESSIONAL EXPERIENCE

Freelance Software & Digital Marketing — Bangladesh 2020 – Present

- Developed and maintained digital solutions for 248+ clients in SaaS, e-commerce and travel, covering web development, SEO and go-to-market (GTM) strategy.
- Helped clients grow their business by up to 45x within 6 months of digitization.

Software Developer Intern — Square InformatiX Ltd. Oct 2022 – Jan 2023

- Developed and unit-tested 6 software components for an Electronic Data Management System, following team coding standards.
- Assisted with database design and query optimization (~35% faster report queries) and resolved 25+ defects, improving reliability and data handling.
- Built user-facing interfaces, maintained system documentation and contributed to requirements gathering.

EDUCATION

BSc in Software Engineering — Yunnan University, China 2019 – 2023
Thesis: ML-Based Movie Recommendation System: Design and Implementation
Relevant areas: Machine Learning, Data Structures & Algorithms, Software Architecture, Software Project Management

CERTIFICATIONS

freeCodeCamp: Python, Machine Learning, Data Science | **HackerRank:** Python, Machine Learning, Data Science

Earlier projects: animal detection, real-time color detection, ASL detection, MySQL course-selection database.